

Master Thesis Supervisor Executive Decision Sheet

Mode-I Adaptive Remeshing Spatial Localization: Diagnosis & Baseline Selection

Date: Thursday, 08 October 2026, 10:00 · Meeting Duration: 45 Minutes

Candidate: Pruthviraja Reddy Vandavagali (Matr. Nr. 68865) · Supervisors: Prof. B. Kiefer, Dr. S. Roth (IMFD)

Governing Directive & Active Phase

“We need to have understood everything related to the first model before we increase complexity.” · Active Phase: MODE1_GATE6B_ACTIVE

1. Problem Observed: Downstream Coarsening in Original Adapted Mesh

In the original Gate-5 adaptive remeshing reproduction (48,329 finite elements), the adapted mesh clustered excessively around the initial notch tip at $(x, y) = (0.50, 0.50)$ mm and coarsened rapidly downstream along the horizontal ligament:

- At initial crack tip $(x = 0.55 \text{ mm})$, resolution was fine with area-equivalent size $h_A = 1.68 \mu\text{m}$.
- Downstream along prospective crack path: $h_A = 4.68 \mu\text{m}$ at $x = 0.65 \text{ mm}$, $h_A = 5.21 \mu\text{m}$ at $x = 0.75 \text{ mm}$, and $h_A = 12.58 \mu\text{m}$ at $x = 0.90 \text{ mm}$ (exceeding the phase-field resolution limit $h \leq l_0/2 = 3.75 \mu\text{m}$ for $l_0 = 7.5 \mu\text{m}$).

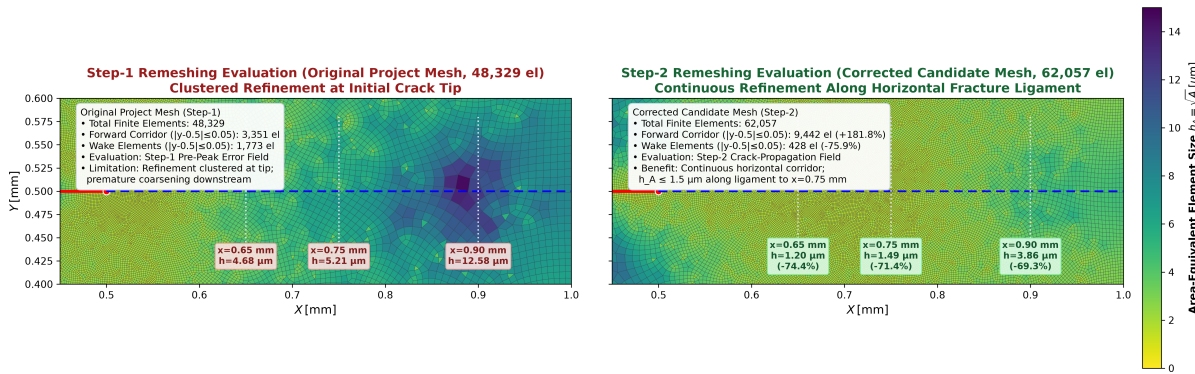
2. Verified Project Root Cause: Abaqus RemeshingRule Step Selection

Controlled CAE diagnostics isolated the exact cause within our Python workflow:

- Step-1 Evaluation:** Evaluating **RemeshingRule** on the Step-1 linear-elastic pre-peak error field (**stepName='Step-1'**) produced stress concentration singularities exclusively at the stationary crack tip (0.5,0.5) mm. Because no crack propagated in Step 1, downstream stress gradients were smooth, causing zero refinement along the ligament.
- Step-2 Evaluation:** Changing **only** the evaluated step to Step-2 crack propagation (**stepName='Step-2'**), with all sizing and indicator parameters unchanged (**MISESERRI**, target 1.0%, $h \in [1.0, 20.0] \mu\text{m}$, **refinementFactor=10**, whole-domain **All_elem**), captures the propagating crack front and extends refinement along the full ligament (62,057 finite elements, 61,646 nodes).

3. Quantitative Evidence: Step-1 vs Step-2 Candidate Comparison

Geometric Metric / Location	Step-1 Mesh (48k)	Step-2 Candidate (62k)	Relative Change	Physical Impact
Forward Crack Corridor ($x \geq 0.5, y - 0.5 \leq 0.05$)	3,351 elements	9,442 elements	+181.8%	2.8× element density along crack path
Crack Wake Region ($x \leq 0.5, y - 0.5 \leq 0.05$)	1,773 elements	428 elements	-75.9%	Suppresses wasteful wake refinement
Direct Ligament Intersects ($y \approx 0.5, x \geq 0.5$)	150 elements	293 elements	+95.3%	Double longitudinal sampling resolution
Median h_A at Station $x = 0.55 \text{ mm}$	1.68 μm	1.89 μm	+12.5%	Sustained within $h \leq 2.0 \mu\text{m}$ ($h/l_0 \leq 0.25$)
Median h_A at Station $x = 0.65 \text{ mm}$	4.68 μm	1.20 μm	-74.4%	Restores fine phase-field resolution
Median h_A at Station $x = 0.75 \text{ mm}$	5.21 μm	1.49 μm	-71.4%	Eliminates coarse downstream barrier
Median h_A at Station $x = 0.90 \text{ mm}$	12.58 μm	3.86 μm	-69.3%	Bounded fine mesh up to boundary edge
Bounded Sizing Compliance ($h \in [1, 20] \mu\text{m}$)	99.47%	99.70%	+0.23%	Strict compliance with sizing constraints



*The original project mesh was generated from the Step-1 pre-peak error field and therefore concentrated refinement around the initial crack-tip region. An isolated change of the Abaqus RemeshingRule evaluation to the Step-2 crack-propagation field, with all sizing parameters unchanged, produced refinement extending along the horizontal ligament. This establishes the project-specific cause of the poor spatial localization. Whether the unpublished Pandey-Kumar implementation used the same step selection cannot be established from the paper.

Figure 1: Side-by-side comparison: Step-1 pre-peak evaluation (48,329 el) vs Step-2 crack-propagation evaluation (62,057 el).

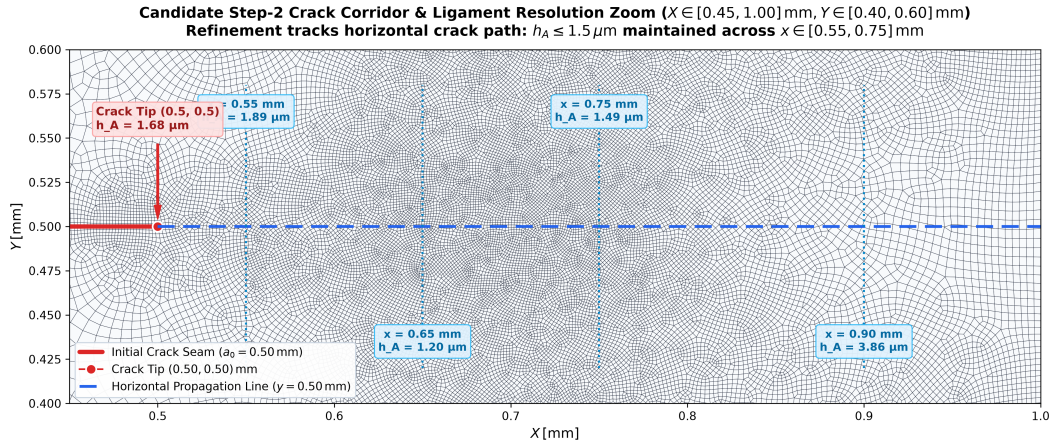


Figure 2: Step-2 crack-corridor zoom ($y \in [0.45, 0.55]$ mm) with downstream cut stations $x \in \{0.55, 0.65, 0.75, 0.90\}$ mm.

4. Epistemological Boundary: What is NOT Known

Epistemological Separation: Project Root Cause vs Published Methodology

- **VERIFIED PROJECT ROOT CAUSE:** Within our Abaqus Python workflow, evaluating `RemeshingRule` on Step-1 vs Step-2 error fields is the verified technical cause that governs tip clustering vs full ligament refinement.
- **UNKNOWN AUTHOR IMPLEMENTATION DETAIL:** Pandey & Kumar (2025, Sec. 4.1) state only that a remeshing rule was created based on MISESERRI at 1%, omitting all script-level `stepName` or frame arguments. We **cannot and do not claim** that Pandey & Kumar used Step 2. Their exact internal selection remains an unresolvable author implementation detail.

5. Mechanical Status & Output-Provenance Audit (Job 1409585.mmater02)

Spatial corridor refinement is **necessary but not sufficient** for scientific adoption:

- **Status:** `ROOT_CAUSE_CANDIDATE_NOT_YET_RECONCILED` (Zero replacement jobs authorized).
- **Mechanical Progression:** Solved 213 increments ($u \rightarrow 0.006554$ mm) with an 87.9% continuous load drop ($F : 0.741 \rightarrow 0.089$ kN), $K_0 = 137.84$ kN/mm ($\Delta = -0.08\%$), and stable horizontal crack extension across **94.0% of ligament** ($x(d \geq 0.5) = 0.9701$ mm, $x(d \geq 0.95) = 0.9513$ mm) with zero local anomalies.
- **Output Provenance Audit:** Traced phase field d (UEL nodal DOF 3) and $\mathcal{H}$ to companion Layer 3 (`All_elem`) `STATEV(1..20)`. The fixed reference deck (1398090.mmater02) requested only `DISP_QUAD` and `N_RP`, omitting `SDV` and mesh `U` fields; its spatial metrics are classified as `NOT_AVAILABLE_FROM_THIS_ODB` / `UNAVAILABLE` rather than false zeros.
- **Boundary Formulation & Bounds:** Decks prescribe displacement boundary conditions only; there is **no explicit phase-field Dirichlet condition**. The weak form implies natural zero-flux ($\partial d / \partial n = 0$). Peak phase field reaches $d_{\max} = 1.0008$ ($+8 \times 10^{-4}$ overshoot above unity); `f42_mixed_uel.for` does not clip d , producing an approximately bounded continuous solution.
- **Diagnostics & Matrix Revision:** MSG scan confirmed **0 negative eigenvalues, 0 singularities, 0 zero-pivots, 0 distorted elements, and 0 warnings**. Termination was triggered solely by $\Delta t < 10^{-8}$ s at final breach. `RIGHT_BOUNDARY_PHASE_FIELD_INTERACTION` is downgraded to `INSUFFICIENT_EVIDENCE` (spatial correlation with advancing crack tip, unproven boundary causation).

6. Formal Supervisor Decision Requested

We request a ruling from the supervisors on the baseline strategy for the Master's thesis:

Option A: Adopt Step-2 Mesh as Baseline for Spatial Localization (Recommended)

Formally designate the 62,057-element Step-2 adapted mesh as the thesis Mode-I Adaptive Baseline for spatial localization, documenting in Chapter 3 that Step-2 evaluation is required to achieve ligament localization under Abaqus native error indicators, while treating final breakthrough termination as a documented numerical open item.

Option B: Retain Step-1 as Literal Reproduction Baseline; Present Step-2 as Project Improvement

Retain the 48,329-element Step-1 mesh in Chapter 3 as the literal reproduction baseline resulting from strict pre-peak error evaluation, and present the 62,057-element Step-2 mesh in Chapter 7 as an author-developed methodology correction that resolves the spatial localization deficit.